

DSA / Algorithms (Abdul Bari) - Day 5

45-day plan mapped to the Algorithms playlist

Asymptotic Notations

Big-O, Omega, Theta and their properties.

Watch these Abdul Bari videos

- > 1.8.1 Asymptotic Notations Big Oh - Omega - Theta #1
- > 1.8.2 Asymptotic Notations #2
- > 1.9 Properties of Asymptotic Notations

Key concepts

- O = upper bound, Omega = lower bound, Theta = tight bound.
- Definitions use constants c and n_0 .
- Properties: transitivity, reflexivity, etc.

Key idea

$f(n) = O(g(n))$ if $f(n) \leq c \cdot g(n)$ for all $n \geq n_0$.

Tasks (1.5h)

- ☐ Watch the 3 videos
- ☐ Prove one Big-O bound
- ☐ List 3 properties

Playlist: youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-029szjTrs_0